REE Professional Subjects: Circuits, Power, Machines, and Measurement

REGISTERED-ELECTRICAL-ENGINEERING · electrical-engineering-professional-subjects · BoardPrep lesson v1

Official examination scope

PRC’s enhanced REE TOS and September 2026 exam program are the official scope references. This guide develops selected electrical principles; current Philippine Electrical Code provisions, grid rules, equipment ratings, and safety requirements must be checked in the applicable current primary text.

Learning objectives

- Interpret a simple calibrated sensor relation and verify its units.
- Explain how circuit, machine, and power-system models connect.
- Recognize where a conceptual calculation stops and current code review begins.

1. Circuit models and steady-state reasoning

A circuit model states element relations, connections, reference directions, and sources. Apply Kirchhoff’s current law at a node and voltage law around a loop, then check whether the solved currents and voltages satisfy each element relation. For sinusoidal steady state, phasors represent magnitude and phase at a common frequency; complex impedance allows resistive and reactive effects to be combined. Keep RMS and peak quantities distinct, use a consistent phase reference, and state the assumed frequency and linear model before interpreting the result.

2. Three-phase power, machines, and system behavior

Three-phase quantities depend on whether the system and measurement are line-to-line or phase values and whether the connection is wye or delta. Power calculations require a consistent voltage, current, power-factor, and RMS convention. Transformers and rotating machines convert electrical and magnetic energy; equivalent circuits help analyze voltage, current, losses, torque, and speed within their assumptions. Power-system analysis adds network topology, operating conditions, demand, and fault behavior. Do not move from a simplified model to a field rating without equipment data and system studies.

3. Instrumentation, feedback, and calibration

A measurement chain includes the physical quantity, sensor, conditioning, conversion, display or recording, and a calibration relationship. Check zero, span, units, range, resolution, uncertainty, and the conditions under which calibration was performed. Feedback compares a measured output with a reference and acts on the difference; stability and response depend on system dynamics and controller settings. For a linear calibration, use the supplied points only within the stated range unless linearity outside it is justified. A reading without an uncertainty or calibration context can overstate precision.

4. Protection, utilization, and current rules

Protection coordinates detection and interruption of abnormal conditions while grounding and bonding manage fault-current paths and touch-risk. Wiring, illumination, and utilization choices depend on load characteristics, equipment, location, and the applicable code edition. Legal and professional responsibilities also matter: verify the current TOS, electrical law, Board issuances, Philippine Electrical Code provisions, and utility or grid requirements before stating a design limit. This lesson teaches how to identify the right check; it does not supply code tables or certify an installation.

Interpolate a sensor calibration

Question: A sensor calibration gives 1 V at 10 units and 3 V at 30 units. Assuming a linear relation over that interval, what voltage corresponds to 20 units?
1. The change is 2 V over 20 units, so the slope is 0.1 V per unit.
2. From 10 to 20 units is 10 units, adding 1 V to the 1 V reference.
3. Check that 20 units is between the two calibration points.
Answer: 2.0 V under the stated linear interpolation assumption.
Check: This does not validate extrapolation outside the calibration interval or the sensor's uncertainty.

Common misconceptions

- A calibration line is valid outside its measured range by default.
- An RMS value and peak value can be substituted without conversion.
- A calculation proves compliance with the current installation code.

Practice patterns

- State reference direction, connection, and RMS convention.
- Keep calibration range and uncertainty visible.
- Name the primary code or system document needed for a compliance claim.

Official scope references

Official scope: Enhanced Tables of Specifications for REE — Annex A — Professional Regulatory Board of Electrical Engineering, Professional Regulation Commission — https://www.prc.gov.ph/sites/default/files/2024-40%20Annex%20A.pdf
Official scope: Program of the September 2026 Registered Electrical Engineers Licensure Examination — Professional Regulation Commission; Professional Regulatory Board of Electrical Engineering — https://www.prc.gov.ph/sites/default/files/Program%20September%202026%20Electrical%20Engineers.pdf

Instructional references

Instructional reference: MIT OpenCourseWare 6.002 Circuits and Electronics — Massachusetts Institute of Technology OpenCourseWare — https://ocw.mit.edu/courses/6-002-circuits-and-electronics-spring-2007/
Instructional reference: MIT OpenCourseWare 6.061 Introduction to Electric Power Systems — Massachusetts Institute of Technology OpenCourseWare — https://ocw.mit.edu/courses/6-061-introduction-to-electric-power-systems-spring-2011/
Instructional reference: MIT OpenCourseWare 6.685 Electric Machines — Massachusetts Institute of Technology OpenCourseWare — https://ocw.mit.edu/courses/6-685-electric-machines-fall-2013/
Instructional reference: University Physics Volume 1 — OpenStax, Rice University — https://openstax.org/details/books/university-physics-volume-1

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